Functional and Structural Brain Age Gaps as Complementary Markers for Cognitive Performance

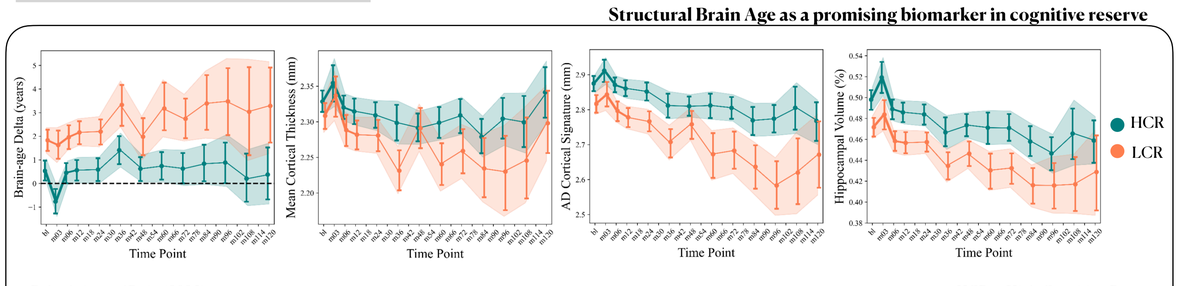
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FRANCE 2030 PROGRAMME DE RECHERCHE SANTÉ NUMÉRIQUE
brain health trajectories

Introduction

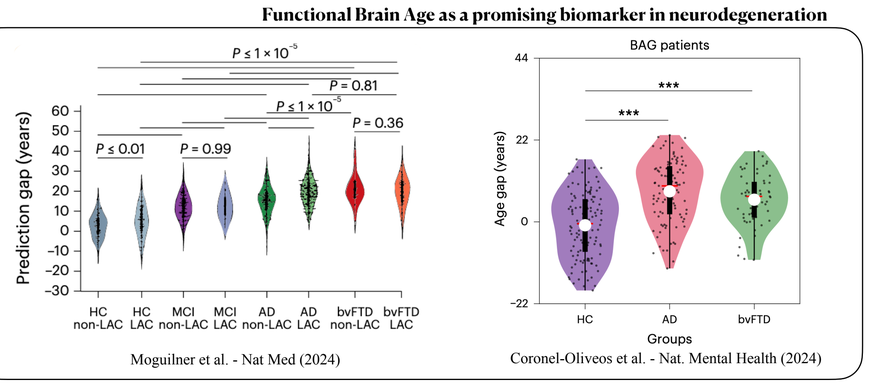
Structural Brain Age as a promising biomarker in cognitive reserve



Beheshti - medRxiv (2025)
Elkana and Beheshti - J. Neurol. (2025)

HCR = High Cognitive Reserve
LCR = Low Cognitive Reserve

Functional Brain Age as a promising biomarker in neurodegeneration

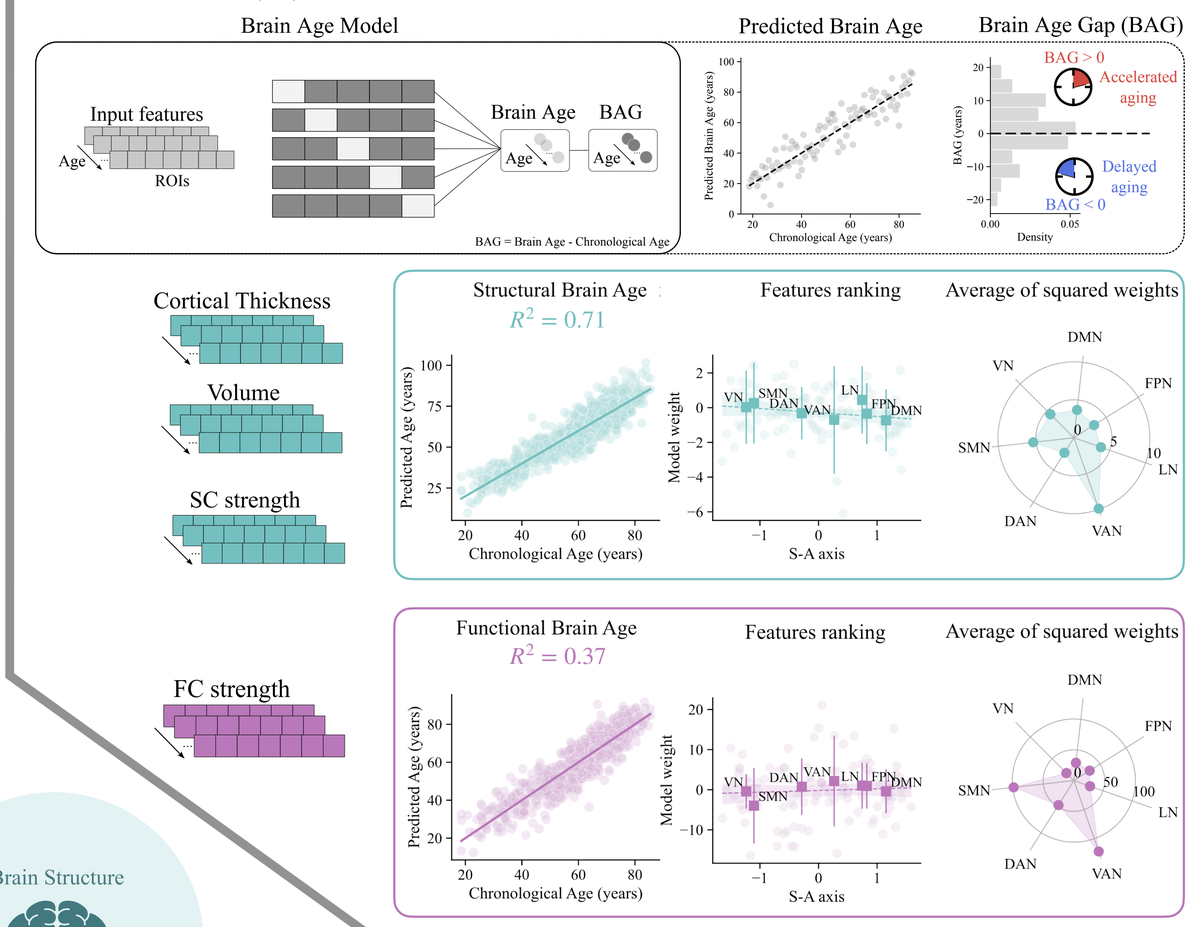


Moguilner et al. - Nat Med (2024)
Coronel-Oliveos et al. - Nat. Mental Health (2024)

"With an increasingly aged population, cognitive impairment is a major health and social issue. Cognitive decline is among the most feared aspects of growing old. It is also the most costly, in terms of the financial, personal and societal burdens." Deary et al. - Br Med Bull (2009)

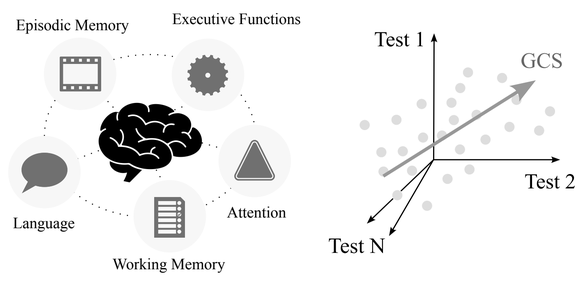
Brain Age Gap (GAP) Framework

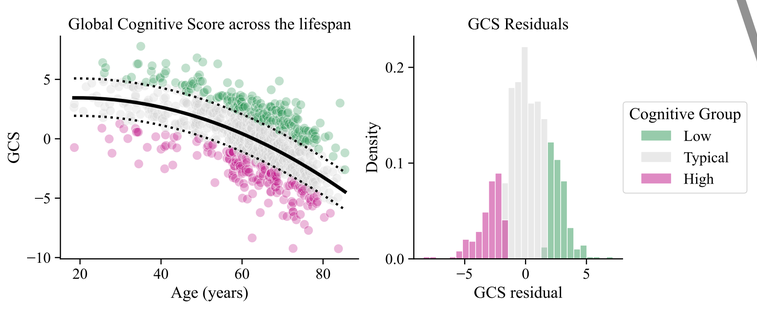
Kalk et al. - Hum. Brain Mapp. (2024)
Cole and Franke - Trends Neurosci. (2012)

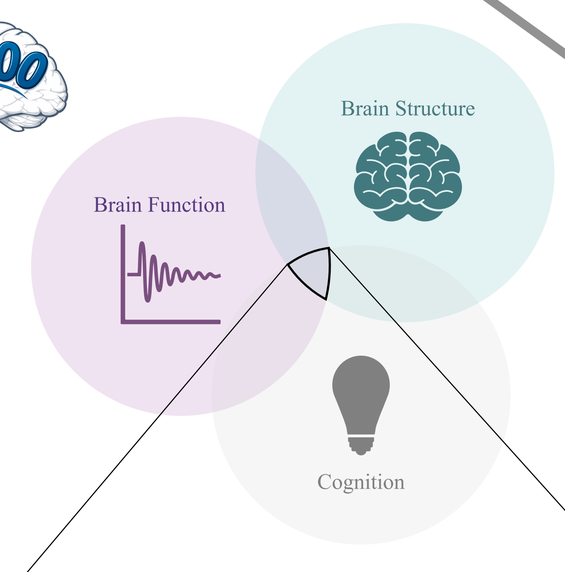


Global Cognitive Score

Caspers et al. - Front Aging Neurosci (2014)

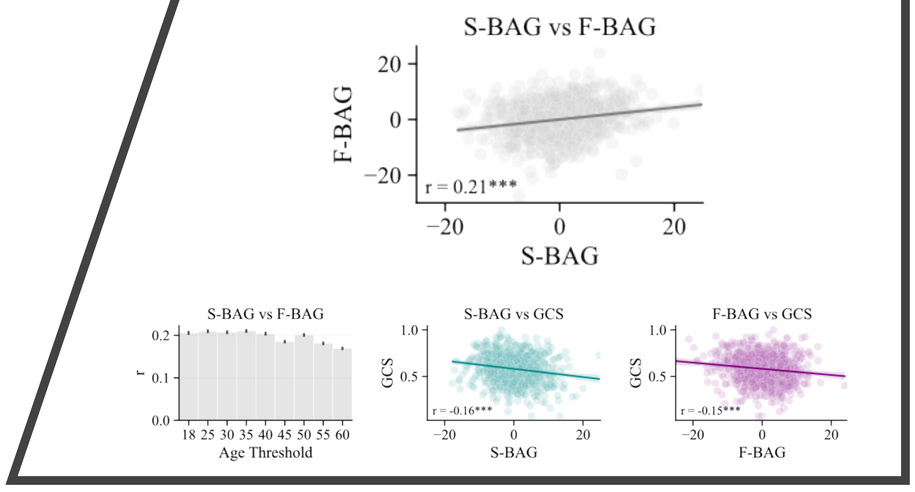






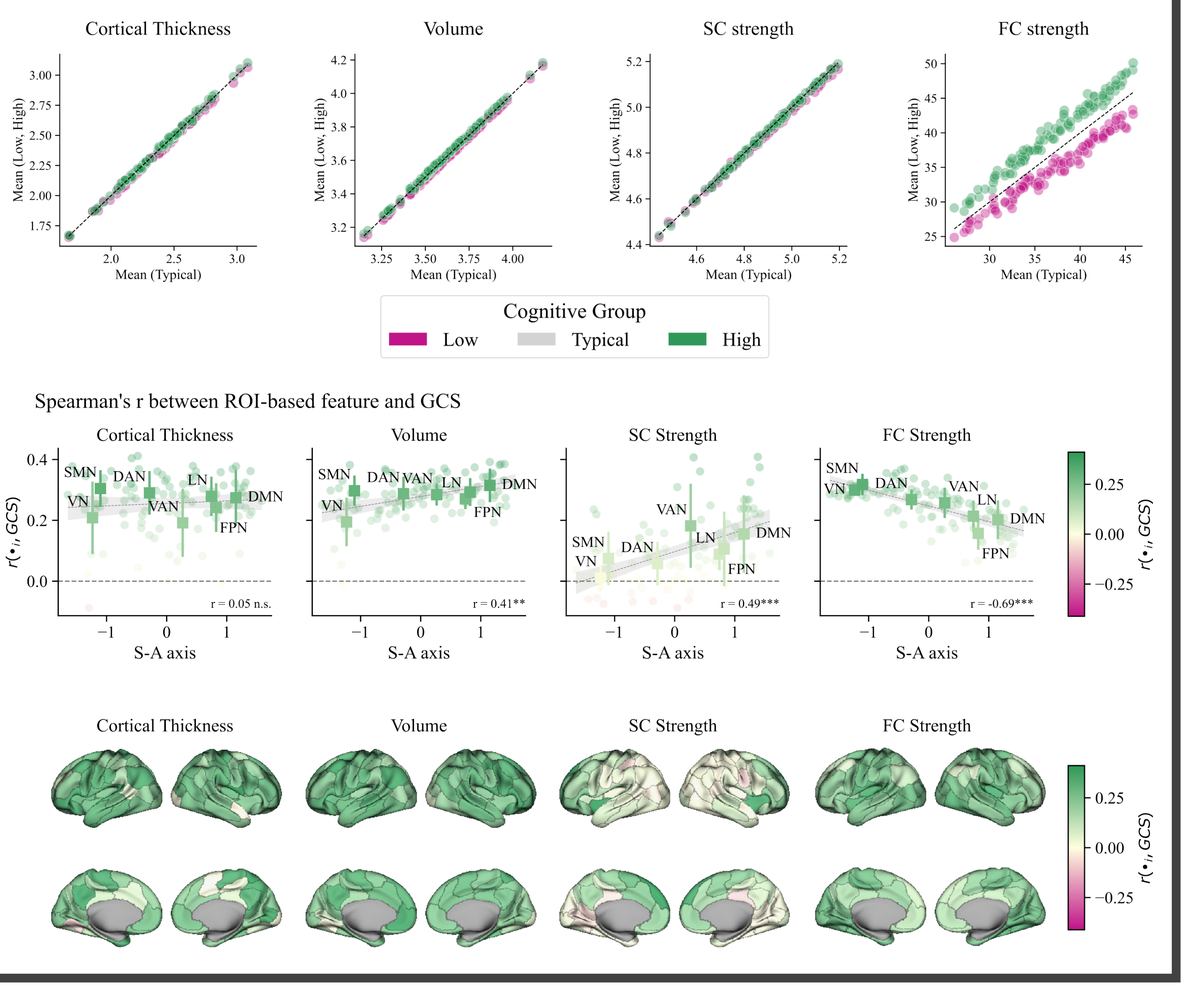
How multimodal brain phenotypes inform the global and domain-specific cognition?

BAGs coupling across the lifespan



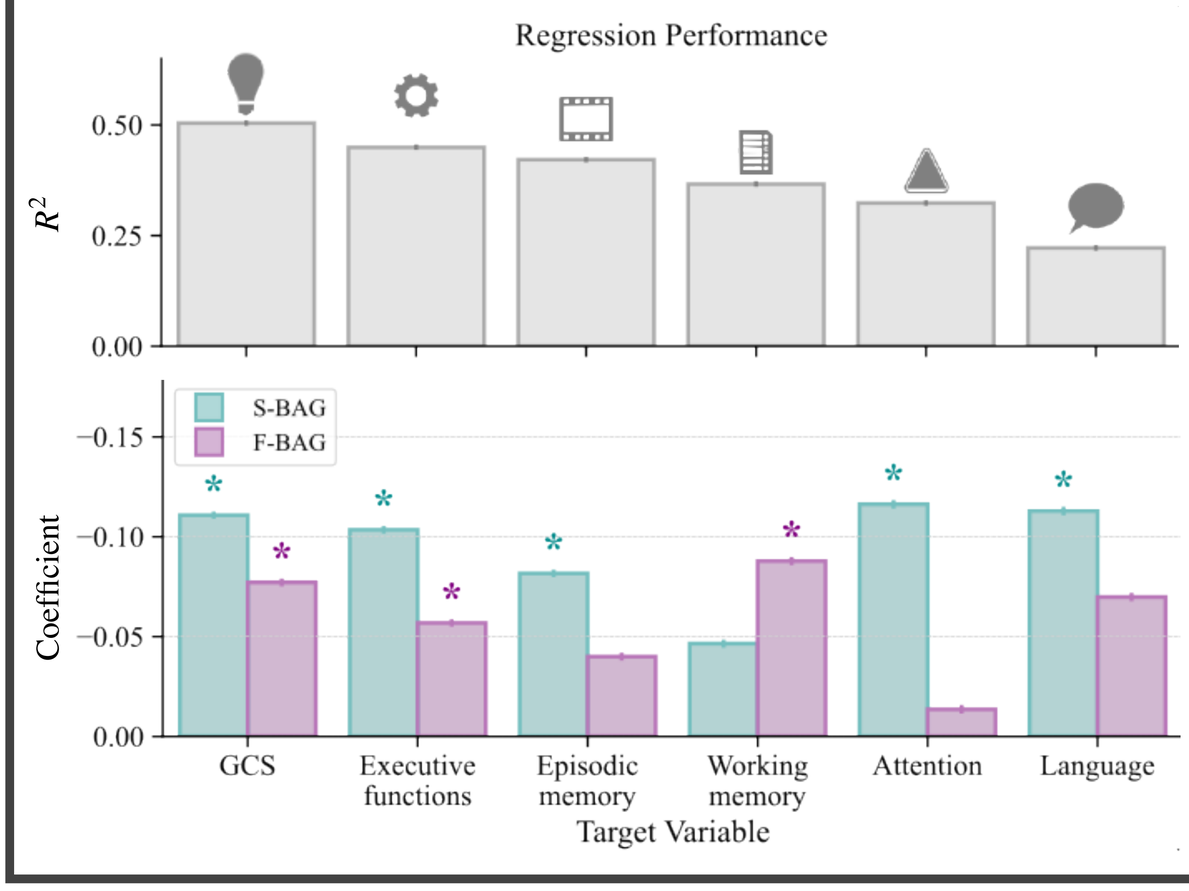
ROI-based features mean by cognitive group

ROI-based features mean across the cognitive groups



BAGs contribution to cognition

Cog Score ~ S-BAG + F-BAG + *covar*



Take-home messages

- Structural and functional brain age models together provide a meaningful subject-level summary of brain information for predicting cognition, supporting a multidimensional view of typical aging.
- The functional marker is as important as the structural marker to predict cognition, even though they are moderately correlated.
- Most accurate prediction was found for global cognition, least accurate prediction for language.
- S-BAG contributes to all domains, whereas F-BAG contributes to global cognition, executive functions and working memory.

Acknowledgements

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